

Computational Finance Report

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Reinforcement Learning for Stock Market Trading Strategy using Q-Learning

This report details the work of our group on developing a stock market trading strategy using Q-Learning. The objective was to train a Reinforcement Learning (RL) agent to learn an optimal trading policy using historical price data and technical indicators. The chosen trading style was not explicitly mentioned.

1. Objective

The primary objective of this project was to **develop a trading strategy using Reinforcement Learning**, specifically by training an RL agent to learn an optimal trading policy. This involved using historical price data and technical indicators as inputs.

2. Data Collection & Preprocessing

The project utilized **historical price data for Apple Inc. (AAPL) obtained from the Yahoo Finance API (yfinance)**. The timeframe was configurable, with the option for daily data. The data acquisition was handled by the `load_data` function.

Technical Indicators

Several technical indicators were implemented, including:

- **Trend Indicators:**
 - SMA (Simple Moving Average) with periods of 20 and 50
- **Momentum Indicators:**
 - RSI (Relative Strength Index), which provides overbought/oversold signals.

Data Normalization & Splitting

The numeric features were scaled between 0 and 1 for model efficiency through **data normalisation**. The data was split temporally into **80% for training and 20% for testing**, handled by the `split_data` function.

3. Defining the RL Components

The Reinforcement Learning framework was defined by its state space, action space, and reward function.

State Space

The state space comprised:

- **Technical Indicators:** SMA (20, 50), RSI.
- **Close Price.**
- **Account Information:** Current balance, shares held, current stock price.

The continuous state space was **discretized** into discrete bins for use with the Q-table. Each discretized state served as a key in the Q-table.

Action Space

The action space was **discrete**:

- **Buy:** Purchase stock.
- **Sell:** Sell held stock.
- **Hold:** Take no action.

Reward Function

The reward function was designed to:

- Provide a **profit-based reward** based on the percentage change in portfolio value.
- Give a **bonus for trades matching technical signals**.
- Apply a **penalty if the portfolio drops by more than 5% (drawdown management)**.
- Include an **inactivity penalty** to discourage prolonged holding without action.

4. RL Algorithm Implementation – Q-Learning

The project implemented **Tabular Q-Learning**. Key components included:

- A **Q-table** to store state-action values, implemented as a nested defaultdict.
- An **epsilon-greedy policy** for balancing exploration and exploitation.
- **Learning rate (α), discount factor (γ), and exploration rate (ϵ).** The learning rate was set to 0.001 and the discount factor to 0.95.
- **Q-value updates** based on the **Bellman Equation**:

- $Q(s,a) = Q(s,a) + \alpha[r + \gamma * \max_{a'} Q(s',a') - Q(s,a)]$.

Implementation Details

- The Q-table maps discretized states to action values, with a default value of 0 for new state-action pairs.
- Continuous states were **discretized** for Q-table use.
- **Action selection** followed an epsilon-greedy policy:
 - With probability ϵ , a random action is chosen (exploration).
 - With probability $1 - \epsilon$, the best-known action is chosen (exploitation).
- The **exploration rate (ϵ)** was gradually reduced over time (exploration rate decay), starting at 1.0 and decreasing to approximately 0.6 over 50 episodes.

5. Training and Evaluation

The dataset was **split into training and testing sets**. The agent learned by maximizing rewards across multiple episodes. The goal was to optimise the agent's policy through repeated interactions with the environment.

Evaluation Metrics

The following metrics were calculated to evaluate the model's performance:

- **Total Profit:** Overall portfolio gain or loss.
- **Sharpe Ratio:** Risk-adjusted return measurement.
- **Max Drawdown:** Largest drop in portfolio value.
- **Win Rate:** Percentage of profitable trades.
- **Cumulative Return:** Total return over time.

Baseline Comparison

The RL strategy's performance was compared with a **Moving Average Strategy** that trades based on predefined moving average crossovers.

6. Visualization and Results

Training Progress

Graphs were generated to show training rewards and portfolio values. The training rewards showed an upward trend over episodes, and the portfolio value grew from \$10,000 with some volatility.

Equity Curves

Equity curves were plotted to compare the performance of the RL strategy, the Moving Average Crossover strategy, and a Buy & Hold strategy.

- **RL Strategy:** Solid blue line, **final return: 7.18%**.
- **MA Crossover:** Orange dashed line, **return: 2.08%**.

Trading Signals on Price Chart

The AAPL price chart was displayed with SMA 20 (green dashed line) and SMA 50 (orange dashed line). Buy signals were indicated by green triangles, and sell signals by red triangles, with entry/exit points concentrated early in the test period.

Q-Learning Progress

The exploration rate showed a gradual drop from 1.0 to approximately 0.6 over 50 episodes.

Performance Metrics Comparison

- **Total Return:** RL strategy outperformed the MA Crossover strategy. The RL strategy had a return of approximately +7.18%, while the MA strategy had a return of approximately +2.08%.
- **Sharpe Ratio:** The RL strategy outperformed the MA Crossover strategy, indicating better risk-adjusted returns.

7. Visualizations and Report

This section provides an overview of the report components.

Model & Training

- **Q-Learning:** A tabular approach with a discretized state space.
- **State Space:** Combines price data, technical indicators, and account info.
- **Action Space:** Discrete – Buy, Sell, Hold.
- **Training Process:** Starts with full exploration ($\epsilon = 1.0$) and gradually shifts to exploitation.
- **Rewards:** Given for profitable trades and indicator-aligned actions.
- **Penalties:** Applied for overtrading and significant drawdowns (>5%).
- **Q-Value Update:** Follows the Bellman equation with a learning rate (α) of 0.001 and a discount factor (γ) of 0.95.

Results Discussion

- **Relative Performance:** The RL strategy yielded a significantly higher return (7.18%) compared to the MA strategy (2.08%).
- **Risk Management & Trading Behavior:** The RL agent executed more trades early in the test period and was more dynamic in response to market shifts than the MA Crossover strategy.

Strengths & Limitations

- **Strengths:**

- Adaptive and market-responsive decision-making.
- Utilises multiple indicators for robust signals.
- **Limitations:**
 - Struggles in strong bear markets.
 - Sensitive to hyperparameters and training duration.

Comparison with Baseline Strategies

- The RL strategy achieved a **5.1% higher return** than the MA Crossover strategy.
- The **RL agent was more active** in trading than the MA Crossover strategy.

